

The Laplace Transform

Signals and Linear Systems

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- **Introduction:** Generalizing the Fourier Transform
- **The Laplace Transform Definition**
- **The Region of Convergence (ROC):**
 - Left-Sided vs. Right-Sided Signals
 - Poles and Zeros
 - Properties of the ROC
- **The Inverse Laplace Transform:**
 - Derivation & Bromwich Contour Integration
 - Finding Inverse via Partial-Fraction Expansion

Introduction and Definition

- **Why general complex exponentials?** We learnt that a broad class of functions can be represented as a sum of complex exponentials $e^{j\omega t}$. Also, complex exponentials are eigenfunctions of LTI systems, i.e. $y(t) = H(j\omega)e^{j\omega t}$ for $x(t) = e^{j\omega t}$.
- Using the properties above, we expressed both input and output of an LTI system as a sum of complex exponentials.
- However, the eigenfunction property holds for any arbitrary complex value $s = \sigma + j\omega$.
- This broader property leads to a generalization of the continuous-time Fourier transform known as the **Laplace Transform**.
- **Key Advantages of the Laplace Transform:**
 - It can analyze broad classes of signals and unstable systems where the Fourier transform does not exist.
 - It provides deeper insights into the **stability or instability** of systems.
 - It maintains powerful algebraic properties, forming an essential set of tools for system analysis, especially **feedback systems**.

- Recall that the response of a Linear Time-Invariant (LTI) system with impulse response $h(t)$ to a complex exponential input $x(t) = e^{st}$ is:

$$\begin{aligned} y(t) &= x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}d\tau \\ &= e^{st} \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau = H(s)e^{st} \end{aligned}$$

- Here, the eigenvalue $H(s)$ corresponding to the eigenfunction e^{st} is given by:

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}dt$$

- For pure imaginary values $s = j\omega$, this integral is simply the **Fourier transform** of $h(t)$.
- For general complex values of s , this is referred to as the **Laplace transform** of the impulse response $h(t)$.

- Extending this beyond impulse responses, we define the Laplace transform of a general signal $x(t)$ as follows:

Laplace Transform

$$X(s) \triangleq \int_{-\infty}^{+\infty} x(t)e^{-st} dt$$

- The complex variable s can be written in rectangular form as $s = \sigma + j\omega$, with σ and ω representing the real and imaginary parts, respectively.
- We sometimes denote the Laplace transform in operator form as $X(s) = \mathcal{L}\{x(t)\}$ and the transform relationship as:

$$x(t) \xleftrightarrow[\mathcal{L}]{\mathcal{L}} X(s)$$

Connection to the Fourier Transform

- When the real part of s is zero ($s = j\omega$), the Laplace transform becomes:

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt = \mathcal{F}\{x(t)\}$$

- The Laplace transform bears a straightforward relationship to the Fourier transform for general $s = \sigma + j\omega$:

$$\begin{aligned} X(\sigma + j\omega) &= \int_{-\infty}^{+\infty} x(t)e^{-(\sigma + j\omega)t} dt \\ &= \int_{-\infty}^{+\infty} [x(t)e^{-\sigma t}]e^{-j\omega t} dt \end{aligned}$$

Key Insight

The Laplace transform of $x(t)$ can be interpreted as the **Fourier transform of** $x(t)e^{-\sigma t}$.

The real exponential weighting $e^{-\sigma t}$ can force the signal into convergence, even if the Fourier transform of $x(t)$ alone does not converge!

Examples of the Laplace Transform

Example 9.1: A Right-Sided Exponential Signal

- Signal: $x(t) = e^{-at}u(t)$.
- Laplace transform:

$$\begin{aligned} X(s) &= \int_0^{\infty} e^{-(s+a)t} dt \\ &= \left[-\frac{1}{s+a} e^{-(s+a)t} \right]_0^{\infty} \end{aligned}$$

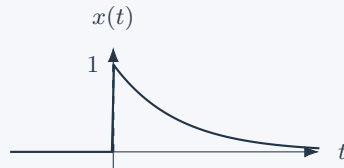
- We evaluate the upper limit:

$$\lim_{t \rightarrow \infty} e^{-(s+a)t} = \lim_{t \rightarrow \infty} e^{-(\sigma+a)t} e^{-j\omega t} = 0 \text{ if } \sigma + a > 0$$

- Hence the integral converges if $\text{Re}\{s\} > -a$.
- Evaluating the lower limit at $t = 0$ yields:

$$X(s) = \frac{1}{s+a}, \quad \text{Re}\{s\} > -a$$

Time Domain $x(t)$



Laplace Domain (s -plane)



Example 9.2: A Left-Sided Exponential Signal

- Signal: $x(t) = -e^{-at}u(-t)$.
- Laplace transform:

$$\begin{aligned} X(s) &= \int_{-\infty}^0 -e^{-(s+a)t} dt \\ &= \left[\frac{1}{s+a} e^{-(s+a)t} \right]_{-\infty}^0 \end{aligned}$$

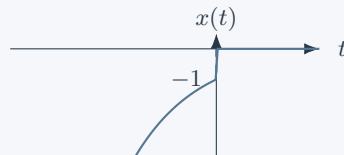
- For convergence at $t \rightarrow -\infty$, exponentials must decay:

$$\sigma + a < 0 \implies \operatorname{Re}\{s\} < -a$$

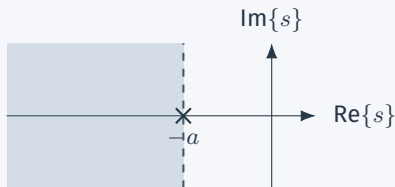
- Evaluating yields the exact same algebraic expression:

$$X(s) = \frac{1}{s+a}, \quad \operatorname{Re}\{s\} < -a$$

Time Domain $x(t)$

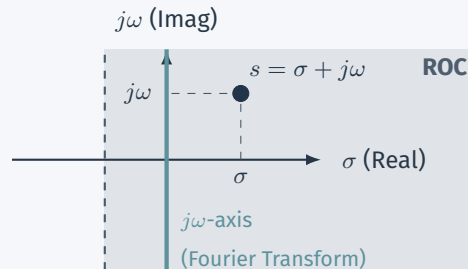


Laplace Domain (s -plane)



The s -plane and Region of Convergence (ROC)

- The Laplace transform depends on the complex variable $s = \sigma + j\omega$.
- We visualize this using a 2D plot called the **s -plane**:
 - **Horizontal axis (σ)**: Real part of s .
 - **Vertical axis ($j\omega$)**: Imaginary part of s .
- **Region of Convergence (ROC)**: The specific set of values in the s -plane for which the integral $X(s)$ converges to a finite value.
 - Unlike Fourier Transform, the ROC is an important part of the Laplace Transform.
 - As we saw in the previous two example, even though the algebraic expression for $X(s)$ was the same, the ROC was different.
- **Relation to Fourier**: The $j\omega$ -axis (where $\sigma = 0$) corresponds to the Fourier transform. If the ROC includes this vertical axis, the Fourier transform of the signal converges!



- Many important LTI systems and signals have Laplace transforms that are **rational functions** of s , meaning they are ratios of two polynomials in s .
- For a signal $x(t)$ with a rational Laplace transform, we can express $X(s)$ as:

$$X(s) = \frac{N(s)}{D(s)}$$

- **Zeros** are the values of s for which $N(s) = 0$ (so $X(s) = 0$). We denote roots of the numerator with an **o** (o) on the s -plane.
- **Poles** are the values of s for which $D(s) = 0$ (so $X(s) \rightarrow \infty$). We denote roots of the denominator with an **x** (x).

Example 9.3: Sum of Two Exponentials

- Signal: $x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$
- Using linearity and Example 9.1:

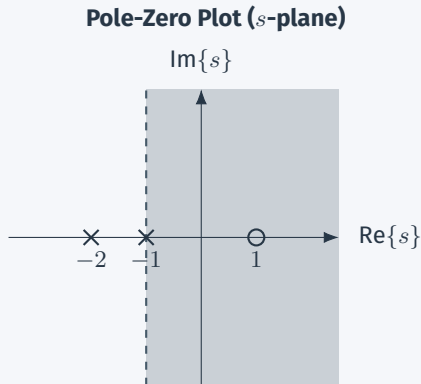
$$3e^{-2t}u(t) \xleftrightarrow{\mathcal{L}} \frac{3}{s+2}, \quad \text{Re}\{s\} > -2$$

$$-2e^{-t}u(t) \xleftrightarrow{\mathcal{L}} \frac{-2}{s+1}, \quad \text{Re}\{s\} > -1$$

- Combined transform:

$$X(s) = \frac{3}{s+2} - \frac{2}{s+1} = \frac{s-1}{(s+1)(s+2)}$$

- **ROC:** Intersection yields $\text{Re}\{s\} > -1$.
- **Zeros:** $s = 1$, **Poles:** $s = -1, -2$



Note: ROC bounded by the right pole ($s = -1$).

Example 9.4: Decay with Oscillation

- Signal: $x(t) = e^{-2t}u(t) + e^{-t}(\cos 3t)u(t)$.
- Expanding $\cos$ via Euler's relation:

$$x(t) = e^{-2t}u(t) + \frac{1}{2}e^{-(1-3j)t}u(t) + \frac{1}{2}e^{-(1+3j)t}u(t)$$

- Laplace transforms of individual terms:

$$X_1(s) = \frac{1}{s+2}, \quad \text{Re}\{s\} > -2$$

$$X_2(s) = \frac{1/2}{s+(1-3j)}, \quad \text{Re}\{s\} > -1$$

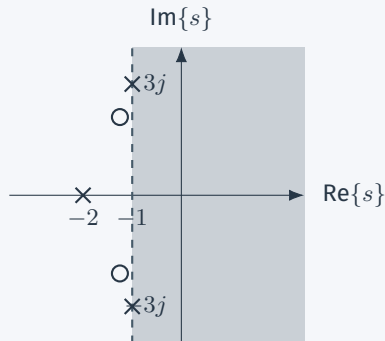
$$X_3(s) = \frac{1/2}{s+(1+3j)}, \quad \text{Re}\{s\} > -1$$

- Combining algebraically gives:

$$X(s) = \frac{2s^2 + 5s + 12}{(s+2)(s^2 + 2s + 10)}, \quad \text{Re}\{s\} > -1$$

- **Poles:** $s = -2, s = -1 \pm 3j$. **Zeros:** $s \approx -1.25 \pm 2.11j$

Pole-Zero Plot (s -plane)



Example 9.5: A Signal with an Impulse

- Signal: $x(t) = \delta(t) - \frac{4}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t)$
- The Laplace transform of the unit impulse converges for the entire s -plane:

$$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-st}dt = 1$$

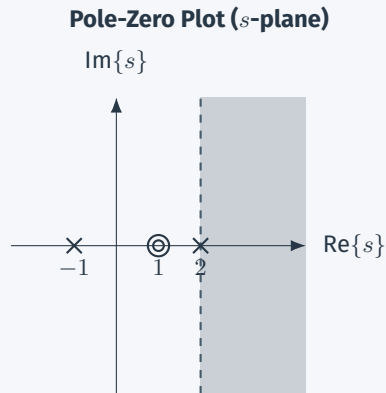
- Thus, the total transform is:

$$X(s) = 1 - \frac{4/3}{s+1} + \frac{1/3}{s-2}$$

- Expanding and keeping common denominator:

$$X(s) = \frac{3s^2 - 6s + 3}{3(s+1)(s-2)} = \frac{(s-1)^2}{(s+1)(s-2)}$$

- **ROC:** Intersection of conditions ($\text{Re}\{s\} > -\infty$, $\text{Re}\{s\} > -1$, $\text{Re}\{s\} > 2$) yields $\text{Re}\{s\} > 2$.



Properties of the Region of Convergence (ROC)

Property 1: The ROC of $X(s)$ consists of strips parallel to the $j\omega$ -axis in the s -plane.

Justification: The Laplace transform converges if $x(t)e^{-st}$ is absolutely integrable. If $\sigma + j\omega$ is in the ROC, then

$$\int_{-\infty}^{\infty} |x(t)e^{-(\sigma+j\omega)t}| dt = \int_{-\infty}^{\infty} |x(t)| e^{-\sigma t} dt < \infty \quad \text{since } |e^{-j\omega t}| = 1$$

Thus, convergence depends entirely on $\sigma = \text{Re}\{s\}$. If a point $\sigma + j\omega$ converges, all points on the vertical line σ must also converge.

Property 2: For rational Laplace transforms, the ROC does not contain any poles.

Justification: By definition, $X(s)$ evaluates to infinity at a pole. Since convergence requires $X(s)$ to be finite, the integral naturally diverges at the exact location of any pole.

Property 3: If $x(t)$ is of finite duration and is absolutely integrable, then the ROC is the entire s -plane.

Justification: Let $x(t) \neq 0$ only in $T_1 < t < T_2$. The absolute integrability integral becomes:

$$\int_{T_1}^{T_2} |x(t)| e^{-\sigma t} dt$$

Since the interval is finite, the maximum value of $e^{-\sigma t}$ over $[T_1, T_2]$ is bounded for any finite σ :

$$\int_{T_1}^{T_2} |x(t)| e^{-\sigma t} dt \leq e^{-\sigma T_{max}} \int_{T_1}^{T_2} |x(t)| dt < \infty \quad \text{where } T_{max} = \begin{cases} T_1 & \text{if } \sigma > 0 \\ T_2 & \text{if } \sigma < 0 \end{cases}$$

This is because if $\sigma > 0$, $e^{-\sigma t}$ decays, so $e^{-\sigma T_1}$ is the largest value of $e^{-\sigma t}$ over $[T_1, T_2]$. If $\sigma < 0$, $e^{-\sigma t}$ grows, so $e^{-\sigma T_2}$ is the largest value of $e^{-\sigma t}$ over $[T_1, T_2]$.

Thus, convergence spans from $-\infty < \sigma < \infty$.

Example 9.6: An Apparent Pole Cancellation

- Consider a finite-duration signal:

$$x(t) = \begin{cases} e^{-at}, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

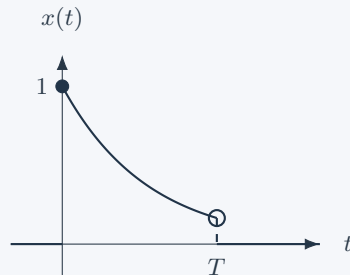
- Evaluating the transform:

$$\begin{aligned} X(s) &= \int_0^T e^{-at} e^{-st} dt \\ &= \frac{1}{s+a} \left[1 - e^{-(s+a)T} \right] \end{aligned}$$

- By **Property 3**, the ROC is the entire s -plane.
- But wait:** There appears to be a pole at $s = -a$. Isn't that a contradiction?
- No! At $s = -a$, both numerator and denominator evaluate to 0. Using L'Hôpital's rule:

$$\begin{aligned} \lim_{s \rightarrow -a} X(s) &= \lim_{s \rightarrow -a} \left[\frac{\frac{d}{ds} (1 - e^{-(s+a)T})}{\frac{d}{ds} (s+a)} \right] \\ &= \lim_{s \rightarrow -a} T e^{-aT} e^{-sT} = T \end{aligned}$$

Time Domain $x(t)$



A finite-duration signal cannot have any true poles because the integrals remain strictly bounded over the finite interval.

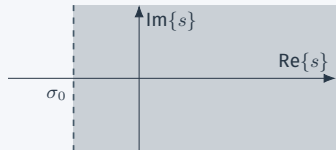
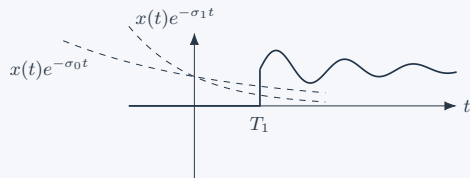
Property 4: If $x(t)$ is right-sided, and if the line $\text{Re}\{s\} = \sigma_0$ is in the ROC, then all values of s for which $\text{Re}\{s\} > \sigma_0$ will also be in the ROC.

Justification: A right-sided signal is zero prior to some finite time T_1 .

If $x(t)e^{-\sigma_0 t}$ converges, then $x(t)e^{-\sigma t}$ will decrease even faster as $t \rightarrow +\infty$ whenever $\sigma_1 > \sigma_0$, as shown in the figure. More formally,

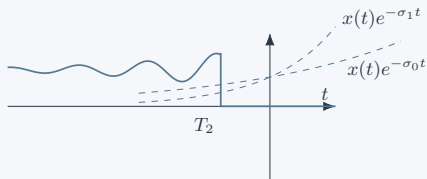
$$\int_{T_1}^{\infty} |x(t)|e^{-\sigma_1 t} dt = \int_{T_1}^{\infty} |x(t)|e^{-\sigma_0 t} e^{-(\sigma_1 - \sigma_0)t} dt \leq e^{-(\sigma_1 - \sigma_0)T_1} \int_{T_1}^{\infty} |x(t)|e^{-\sigma_0 t} dt < \infty$$

As $e^{-(\sigma_1 - \sigma_0)T_1}$ and $\int_{T_1}^{\infty} |x(t)|e^{-\sigma_0 t} dt$ are finite.

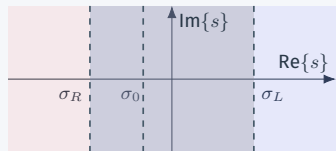
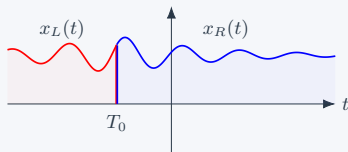


Property 5: If $x(t)$ is left-sided, and if the line $\text{Re}\{s\} = \sigma_0$ is in the ROC, then all values of s for which $\text{Re}\{s\} < \sigma_0$ will also be in the ROC.

Justification: Analogous to Property 4, the signal extends to $t \rightarrow -\infty$. For convergence, we need exponential compounding to die out in the negative time direction, which occurs when $\sigma_1 < \sigma_0$.



Property 6: If $x(t)$ is two-sided, and if the line $\text{Re}\{s\} = \sigma_0$ is in the ROC, then the ROC will consist of a strip in the s -plane that includes the line $\text{Re}\{s\} = \sigma_0$.



- $x(t)$ can be divided at any arbitrary T_0 into $x_L(t)$ (red) and $x_R(t)$ (blue) such that $x(t) = x_L(t) + x_R(t)$.
- As $\mathcal{L}\{x(t)\} = \mathcal{L}\{x_L(t)\} + \mathcal{L}\{x_R(t)\}$ converges, $\mathcal{L}\{x_L(t)\}$ and $\mathcal{L}\{x_R(t)\}$ must converge with the ROC $\text{Re}\{s\} < \sigma_L$ (red) and $\text{Re}\{s\} > \sigma_R$ (blue) respectively according to the property 4 and 5 for some σ_L and σ_R .

- The ROC of $x(t)$ is the intersection of the ROCs of $x_L(t)$ and $x_R(t)$, which is the region $\sigma_R < \text{Re}\{s\} \leq \sigma_L$.
- Since σ_0 is in the ROC, we must have $\sigma_R < \sigma_0 \leq \sigma_L$.
- Thus, the ROC is a strip in the s -plane that includes the line $\text{Re}\{s\} = \sigma_0$.

Example 9.7: A Two-Sided Exponential Signal

- Signal: $x(t) = e^{-b|t|}$, for $b > 0$.
- Since this is a two-sided signal, we can express it as the sum of a right-sided and a left-sided signal:

$$x(t) = e^{-bt}u(t) + e^{bt}u(-t)$$

- From previous examples, the individual transforms are:

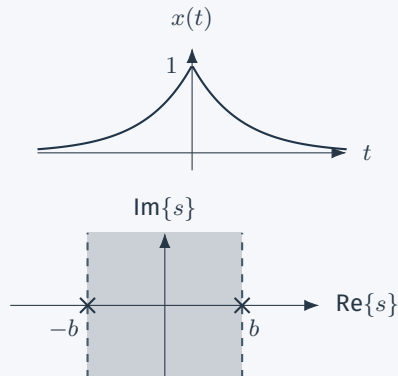
$$e^{-bt}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+b}, \quad \text{Re}\{s\} > -b$$

$$e^{bt}u(-t) \xleftrightarrow{\mathcal{L}} \frac{-1}{s-b}, \quad \text{Re}\{s\} < b$$

- For the total signal to have a Laplace transform, the two ROCs must overlap. Thus, we require $-b < \text{Re}\{s\} < b$.
- If $b \leq 0$, there is no common ROC, and the Laplace transform does not exist.
- For $b > 0$, combining the transforms gives:

$$X(s) = \frac{1}{s+b} - \frac{1}{s-b} = \frac{-2b}{s^2 - b^2}$$

- **ROC:** $-b < \text{Re}\{s\} < b$. **Poles:** $s = \pm b$.



Property 7: If the Laplace transform $X(s)$ is rational, then its ROC is bounded by poles or extends to infinity. In addition, no poles are contained in the ROC.

Justification: Rational transforms are combinations of exponentials (we will see this soon). And we saw already that the ROC of an exponential is bounded by a pole (e.g. $e^{-at}u(t)$ has a pole at $-a$ and the ROC extends from $-a$ to ∞).

As the ROC of a linear combination of signals is the intersection of the ROCs of the individual signals, the ROC of a rational transform is bounded by poles or extends to infinity.

Property 8: If $X(s)$ is rational, then:

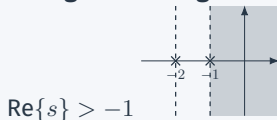
- If $x(t)$ is right-sided, the ROC is the region to the **right of the rightmost pole**.
- If $x(t)$ is left-sided, the ROC is the region to the **left of the leftmost pole**.

Justification: Combination of property 4 and 5 with property 7.

Example 9.8: One Equation, Three Signals

- Consider the rational transform: $X(s) = \frac{1}{(s+1)(s+2)}$
- Without specifying the ROC, this maps to **three** possible time-domain signals depending on the bounding poles ($s = -1, -2$):

Right-Sided Signal



Left-Sided Signal



Two-Sided Signal



The Inverse Laplace Transform

- Recall that we interpreted the Laplace transform of $x(t)$ as the Fourier transform of the exponentially weighted signal $x(t)e^{-\sigma t}$:

$$X(\sigma + j\omega) = \mathcal{F}\{x(t)e^{-\sigma t}\} = \int_{-\infty}^{\infty} x(t)e^{-\sigma t}e^{-j\omega t}dt$$

- We can apply the **Inverse Continuous-Time Fourier Transform** to recover the weighted signal $x(t)e^{-\sigma t}$:

$$x(t)e^{-\sigma t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega)e^{j\omega t}d\omega$$

- Multiplying both sides by $e^{\sigma t}$ yields:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega)e^{(\sigma + j\omega)t}d\omega$$

- Let's perform a change of variables in the integration. We substitute $s = \sigma + j\omega$.
- Since σ is treated as constant during this integration, the differential becomes $ds = j d\omega$, or $d\omega = \frac{ds}{j}$.
- The limits of integration for ω are $-\infty$ to $+\infty$, which correspond to s running from $\sigma - j\infty$ to $\sigma + j\infty$.
- σ must be chosen within the ROC of $X(s)$, otherwise $X(s)$ is not defined.

Inverse Laplace Transform

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{st} ds$$

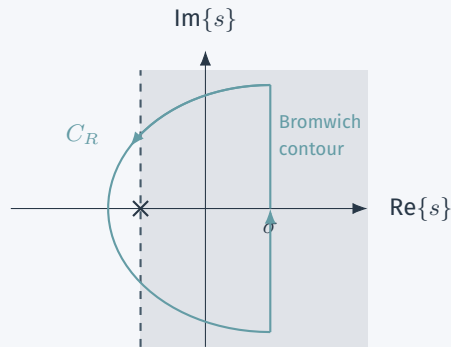
where σ can be chosen anywhere within the ROC of $X(s)$.

Evaluating the Inverse: Bromwich Integral

- The equation above is known as the **Bromwich Integral** or the Mellin inversion formula.
- This involves a **contour integration** in the complex s -plane along a straight vertical line $\text{Re}\{s\} = \sigma$.
- In practice, this contour is closed along an infinite semicircle to the left or right (depending on $t > 0$ or $t < 0$), allowing the use of the **Cauchy Residue Theorem**:

$$\oint_C f(s)ds = 2\pi j \sum \text{Residues enclosed by } C$$

- However, directly evaluating Cauchy residues via contour integrals is mathematically a bit tough.
- If you are still interested you can check this video: <https://youtu.be/r68zKdyIwVo>



Contour closed to the left for $t > 0$, enclosing poles.

- Sometimes we may not have an analytical inverse, or we only need a numerical approximation of $x(t)$.
- Recall that the inverse Laplace transform can be written as an integration over ω :

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{(\sigma + j\omega)t} d\omega$$

- We can approximate this continuous integral with a discrete Riemann sum by taking small frequency steps $\Delta\omega$:

$$x(t) \approx \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(\sigma + jk\Delta\omega) e^{(\sigma + jk\Delta\omega)t} \Delta\omega$$

Numerical Approx. Inverse Laplace

$$x(t) \approx \frac{\Delta\omega}{2\pi} \sum_{k=-\infty}^{\infty} X(\sigma + jk\Delta\omega) e^{(\sigma + jk\Delta\omega)t}$$

σ must still be chosen within the ROC.

In many real-world problems we are dealing with rational Laplace transforms.

- Instead of using the formal contour integration, we heavily rely on **Partial-Fraction Expansion**.
- For rational Laplace transforms $X(s) = \frac{N(s)}{D(s)}$ with simple poles at $s = -a_i$, we expand carefully:

$$X(s) = \sum_{i=1}^M \frac{A_i}{s + a_i}$$

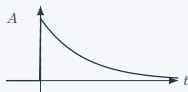
- Because the Laplace transform is linear, we simply invert each first-order term $\frac{A_i}{s + a_i}$ individually.
- **Importantly**, to invert a term, we must know its ROC. There are two distinct time-domain signals for every term $\frac{A_i}{s + a_i}$ depending on the assumed ROC.

If the fractional term is $\frac{A}{s+a}$:

ROC is to the right ($\text{Re}\{s\} > -a$)

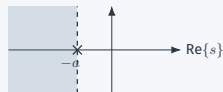


$$\frac{A}{s+a} \xleftrightarrow{\mathcal{L}} Ae^{-at}u(t)$$



(Right-sided signal)

ROC is to the left ($\text{Re}\{s\} < -a$)



$$\frac{A}{s+a} \xleftrightarrow{\mathcal{L}} -Ae^{-at}u(-t)$$



(Left-sided signal)

- Let our transform be the system from Example 9.8:

$$X(s) = \frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

- Using the cover-up method or equivalent algebra:

$$A = (s+1)X(s)|_{s=-1} = \frac{1}{s+2} \Big|_{s=-1} = 1$$

$$B = (s+2)X(s)|_{s=-2} = \frac{1}{s+1} \Big|_{s=-2} = -1$$

- Thus, the expanded form is ALWAYS:

$$X(s) = \frac{1}{s+1} - \frac{1}{s+2}$$

Example 9.9: Inverting a Right-Sided ROC

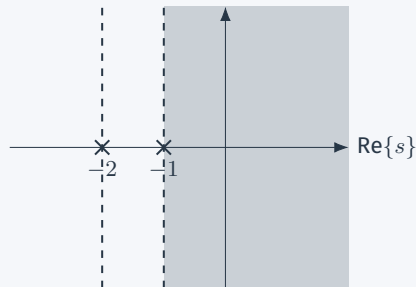
- Suppose the given ROC is $\mathbf{Re}\{s\} > -1$.
- By evaluating the overall ROC against each term's pole, we can find the individual ROCs:

$$\text{Pole at } s = -1 : \text{ ROC is to the right. } \Rightarrow \frac{1}{s+1} \xleftrightarrow{\mathcal{L}} +e^{-t}u(t)$$

$$\text{Pole at } s = -2 : \text{ ROC is to the right. } \Rightarrow \frac{1}{s+2} \xleftrightarrow{\mathcal{L}} +e^{-2t}u(t)$$

Resulting inverse transform:

$$x(t) = (e^{-t} - e^{-2t}) u(t)$$



Example 9.10: Inverting a Left-Sided ROC

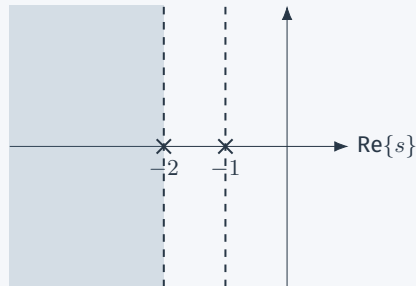
- Now suppose the given ROC is $\text{Re}\{s\} < -2$.
- By evaluating the overall ROC against each term's pole, we can find the individual ROCs:

$$\text{Pole at } s = -1 : \text{ ROC is to the left. } \Rightarrow \frac{1}{s+1} \xleftrightarrow{\mathcal{L}} -e^{-t}u(-t)$$

$$\text{Pole at } s = -2 : \text{ ROC is to the left. } \Rightarrow \frac{1}{s+2} \xleftrightarrow{\mathcal{L}} -e^{-2t}u(-t)$$

Resulting inverse transform:

$$x(t) = (-e^{-t} + e^{-2t}) u(-t)$$



Example 9.11: Inverting a Two-Sided ROC

- Finally, suppose the ROC is $-2 < \text{Re}\{s\} < -1$.
- By evaluating the overall ROC against each term's pole, we can find the individual ROCs:

$$\text{Pole at } s = -1 : \text{ ROC is to the left. } \Rightarrow \frac{1}{s+1} \xleftrightarrow{\mathcal{L}} -e^{-t}u(-t)$$

$$\text{Pole at } s = -2 : \text{ ROC is to the right. } \Rightarrow \frac{1}{s+2} \xleftrightarrow{\mathcal{L}} +e^{-2t}u(t)$$

Resulting inverse transform:

$$x(t) = -e^{-t}u(-t) - e^{-2t}u(t)$$

